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Probability and Statistics Lab (DA-1)

Question: For each of the three data sets provided below, calculate the following statistical measures: Mean (using direct method, assumed mean method, step deviation method), Median, Mode, Range, coefficient of range, Quartile Deviation, coefficient of Q.D, Mean Deviation from mean, coefficient of M.D, Mean Deviation from median, coefficient of M.D, Standard Deviation (Direct, Assumed Mean Method, Step Deviation Method).

1. A science teacher administered a chemistry quiz to a class of 25 students. The maximum possible score was 20 points. The teacher recorded the raw scores for each student to analyse the class's performance.

12, 15, 18, 14, 16, 12, 19, 15, 13, 17, 15, 11, 14, 18, 20, 15, 13, 16, 12, 19, 14, 10, 17, 15, 18

CODE:-

```
DA1.R* x
Source on Save
Run
Source

1 #Question1:
2 x <- c(12,15,18,14,16,12,19,15,13,17,15,11,14,18,20,
3        15,13,16,12,19,14,10,17,15,18)
4 #MEAN (Three Methods)
5 #(a) Direct Method
6 mean_x <- sum(x) / length(x)
7 mean_x
8 #(b) Assumed Mean Method
9 A <- 15 # assumed mean
10 d <- x - A
11 mean_assumed <- A + sum(d) / length(x)
12 mean_assumed
13 #(c) Step Deviation Method
14 h <- 1
15 u <- (x - A) / h
16 mean_step <- A + (sum(u) / length(x)) * h
17 mean_step
18 #MEDIAN
19 median_x <- median(x)
20 median_x
21 #MODE
22 freq <- table(x)
23 mode_x <- as.numeric(names(freq[freq == max(freq)]))
24 mode_x
25 #RANGE & COEFFICIENT OF RANGE
26 range_x <- max(x) - min(x)
27 range_x
28 coef_range <- (max(x) - min(x)) / (max(x) + min(x))
29 coef_range
30 #QUARTILE DEVIATION & COEFFICIENT
31 Q1 <- as.numeric(quantile(x, 0.25))
32 Q3 <- as.numeric(quantile(x, 0.75))
33 QD <- (Q3 - Q1) / 2
34 QD
35 coef_QD <- (Q3 - Q1) / (Q3 + Q1)
36 coef_QD
37 #MEAN DEVIATION FROM MEAN & COEFFICIENT
38 MD_mean <- sum(abs(x - mean_x)) / length(x)
39 MD_mean
```

CONSOLE:-

Source

Console

Terminal x

Background Jobs x

R 4.5.2 · ~/new da/

```
> #Question1:
> x <- c(12,15,18,14,16,12,19,15,13,17,15,11,14,18,20,
+       15,13,16,12,19,14,10,17,15,18)
> #MEAN (Three Methods)
> #(a) Direct Method
> mean_x <- sum(x) / length(x)
> mean_x
[1] 15.12
> #(b) Assumed Mean Method
> A <- 15 # assumed mean
> d <- x - A
> mean_assumed <- A + sum(d) / length(x)
> mean_assumed
[1] 15.12
> #(c) Step Deviation Method
> h <- 1
> u <- (x - A) / h
> mean_step <- A + (sum(u) / length(x)) * h
> mean_step
[1] 15.12
> #MEDIAN
> median_x <- median(x)
> median_x
[1] 15
> #MODE
> freq <- table(x)
> mode_x <- as.numeric(names(freq[freq == max(freq)]))
> mode_x
[1] 15
> #RANGE & COEFFICIENT OF RANGE
> range_x <- max(x) - min(x)
> range_x
[1] 10
> coef_range <- (max(x) - min(x)) / (max(x) + min(x))
> coef_range
[1] 0.3333333
> #QUARTILE DEVIATION & COEFFICIENT
> Q1 <- as.numeric(quantile(x, 0.25))
> Q3 <- as.numeric(quantile(x, 0.75))
> Q1
[1] 14
> Q3
[1] 18
```

```
Source
Console Terminal Background Jobs
R • R 4.5.2 • ~/new da/
> #QUARTILE DEVIATION & COEFFICIENT
> Q1 <- as.numeric(quantile(x, 0.25))
> Q3 <- as.numeric(quantile(x, 0.75))
> QD <- (Q3 - Q1) / 2
> QD
[1] 2
> coef_QD <- (Q3 - Q1) / (Q3 + Q1)
> coef_QD
[1] 0.1333333
> #MEAN DEVIATION FROM MEAN & COEFFICIENT
> MD_mean <- sum(abs(x - mean_x)) / length(x)
> MD_mean
[1] 2.144
> coef_MD_mean <- MD_mean / mean_x
> coef_MD_mean
[1] 0.1417989
> #MEAN DEVIATION FROM MEDIAN & COEFFICIENT
> MD_median <- sum(abs(x - median_x)) / length(x)
> MD_median
[1] 2.12
> coef_MD_median <- MD_median / median_x
> coef_MD_median
[1] 0.1413333
> #STANDARD DEVIATION (Three Methods)
> #(a) Direct Method
> SD_direct <- sqrt(sum((x - mean_x)^2) / length(x))
> SD_direct
[1] 2.627851
> #(b) Assumed Mean Method
> SD_assumed <- sqrt(sum(d^2) / length(x) - (sum(d) /
+ length(x))^2)
> SD_assumed
[1] 2.627851
> #(c) Step Deviation Method
> SD_step <- sqrt( (sum(u^2) / length(x)) - (sum(u) /
+ length(x))^2 ) * h
> SD_step
[1] 2.627851
> |
```

OUTPUT:-

Environment	History	Connections	Tutorial
Import Dataset	16 MiB		
R	Global Environment		
Values			
A	15		
coef_MD_mean	0.141798941798942		
coef_MD_median	0.141333333333333		
coef_QD	0.133333333333333		
coef_range	0.333333333333333		
d	num [1:25] -3 0 3 -1 1 -3 4 0 -2 2 ...		
freq	'table' int [1:11(1d)] 1 1 3 2 3 5 2 2 3 2 ...		
h	1		
MD_mean	2.144		
MD_median	2.12		
mean_assumed	15.12	MD_mean (numeric , 56 bytes)	
mean_score	15.12		
mean_step	15.12		
mean_x	15.12		
median_score	15		
median_x	15		
mode_score	15		
mode_x	15		
n	25L		
Q1	13		
Q3	17		
QD	2		
range_score	10		
range_x	10		
scores	num [1:25] 12 15 18 14 16 12 19 15 13 17 ...		
SD_assumed	2.62785083290509		
SD_direct	2.62785083290509		
SD_step	2.62785083290509		
u	num [1:25] -3 0 3 -1 1 -3 4 0 -2 2 ...		
x	num [1:25] 12 15 18 14 16 12 19 15 13 17 ...		
Functions			
Files	Plots	Packages	Help Viewer Presentation

Que 1 (code part):-

#Question1:

```
x <- c(12,15,18,14,16,12,19,15,13,17,15,11,14,18,20,
15,13,16,12,19,14,10,17,15,18)
```

#MEAN (Three Methods)

#(a) Direct Method

```
mean_x <- sum(x) / length(x)
```

```

mean_x
#(b) Assumed Mean Method
A <- 15 # assumed mean
d <- x - A
mean_assumed <- A + sum(d) / length(x)
mean_assumed
#(c) Step Deviation Method
h <- 1
u <- (x - A) / h
mean_step <- A + (sum(u) / length(x)) * h
mean_step
#MEDIAN
median_x <- median(x)
median_x
#MODE
freq <- table(x)
mode_x <- as.numeric(names(freq[freq == max(freq)]))
mode_x
#RANGE & COEFFICIENT OF RANGE
range_x <- max(x) - min(x)
range_x
coef_range <- (max(x) - min(x)) / (max(x) + min(x))
coef_range#QUARTILE DEVIATION & COEFFICIENT
Q1 <- as.numeric(quantile(x, 0.25))
Q3 <- as.numeric(quantile(x, 0.75))
QD <- (Q3 - Q1) / 2
QD
coef_QD <- (Q3 - Q1) / (Q3 + Q1)
coef_QD
#MEAN DEVIATION FROM MEAN & COEFFICIENT
MD_mean <- sum(abs(x - mean_x)) / length(x)
MD_mean
coef_MD_mean <- MD_mean / mean_x
coef_MD_mean

```

```

#MEAN DEVIATION FROM MEDIAN & COEFFICIENT
MD_median <- sum(abs(x - median_x)) / length(x)
MD_median
coef_MD_median <- MD_median / median_x
coef_MD_median
#STANDARD DEVIATION (Three Methods)
#(a) Direct Method
SD_direct <- sqrt(sum((x - mean_x)^2) / length(x))
SD_direct
#(b) Assumed Mean Method
SD_assumed <- sqrt(sum(d^2) / length(x) - (sum(d) /
length(x))^2)
SD_assumed
#(c) Step Deviation Method
SD_step <- sqrt( (sum(u^2) / length(x)) - (sum(u) /
length(x))^2 ) * h
SD_step

```

QUE-2

2. A local coffee shop surveyed 40 regular customers to find out how many times they visited the shop in the last week. The results are organized into the frequency table below.

X (No. of Visits) 0 1 2 3 4 5

f (No. of Customers) 4 12 11 7 4 2

CODE:-

```
DA1.R* x
Source on Save
Run
Source

1 #Question 2:
2 x <- c(0,1,2,3,4,5)
3 f <- c(4,12,11,7,4,2)
4 data <- rep(x, f)
5 length(data)
6 #MEAN
7 #(A)DIRECT METHOD
8 mean_x <- sum(x * f) / sum(f)
9 mean_x
10 #(b) Assumed Mean Method
11 A <- 2 # assumed mean
12 d <- x - A
13 mean_assumed <- A + sum(f * d) / sum(f)
14 mean_assumed
15 #(c) Step Deviation Method
16 h <- 1
17 u <- (x - A) / h
18 mean_step <- A + (sum(f * u) / sum(f)) * h
19 mean_step
20 #MEDIAN
21 median_x <- median(data)
22 median_x
23 #Mode
24 mode_x <- x[f == max(f)]
25 mode_x
26 #RANGE & COEFFICIENT OF RANGE
27 range_x <- max(x) - min(x)
28 range_x
29 coef_range <- (max(x) - min(x)) / (max(x) + min(x))
30 coef_range
31 #QUARTILE DEVIATION (Q.D.) & COEFFICIENT
32 Q1 <- as.numeric(quantile(data, 0.25))
33 Q3 <- as.numeric(quantile(data, 0.75))
34 QD <- (Q3 - Q1) / 2
35 QD
36 coef_QD <- (Q3 - Q1) / (Q3 + Q1)
37 coef_QD
38 #MEAN DEVIATION ABOUT MEAN & COEFFICIENT
39 MD_mean <- sum(abs(data - mean_x)) / length(data)

59:8 (Top Level) ↕ R Script ↕
```

CONSOLE:-

Source



Console

Terminal

Background Jobs



R · R 4.5.2 · ~/new da/



```
> #Question 2:
> x <- c(0,1,2,3,4,5)
> f <- c(4,12,11,7,4,2)
> data <- rep(x, f)
> length(data)
[1] 40
> #MEAN
> #(A)DIRECT METHOD
> mean_x <- sum(x * f) / sum(f)
> mean_x
[1] 2.025
> #(b) Assumed Mean Method
> A <- 2 # assumed mean
> d <- x - A
> mean_assumed <- A + sum(f * d) / sum(f)
> mean_assumed
[1] 2.025
> #(c) Step Deviation Method
> h <- 1
> u <- (x - A) / h
> mean_step <- A + (sum(f * u) / sum(f)) * h
> mean_step
[1] 2.025
> #MEDIAN
> median_x <- median(data)
> median_x
[1] 2
> #Mode
> mode_x <- x[f == max(f)]
> mode_x
[1] 1
> #RANGE & COEFFICIENT OF RANGE
> range_x <- max(x) - min(x)
> range_x
[1] 5
> coef_range <- (max(x) - min(x)) / (max(x) + min(x))
> coef_range
[1] 1
> #QUARTILE DEVIATION (O.D.) & COEFFICIENT
```

```
Source
Console Terminal x Background Jobs x
R 4.5.2 · ~/new da/
> #QUARTILE DEVIATION (Q.D.) & COEFFICIENT
> Q1 <- as.numeric(quantile(data, 0.25))
> Q3 <- as.numeric(quantile(data, 0.75))
> QD <- (Q3 - Q1) / 2
> QD
[1] 1
> coef_QD <- (Q3 - Q1) / (Q3 + Q1)
> coef_QD
[1] 0.5
> #MEAN DEVIATION ABOUT MEAN & COEFFICIENT
> MD_mean <- sum(abs(data - mean_x)) / length(data)
> MD_mean
[1] 1.03375
> coef_MD_mean <- MD_mean / mean_x
> coef_MD_mean
[1] 0.5104938
> #MEAN DEVIATION ABOUT MEDIAN & COEFFICIENT
> MD_median <- sum(abs(data - median_x)) / length(data)
> MD_median
[1] 1.025
> coef_MD_median <- MD_median / median_x
> coef_MD_median
[1] 0.5125
> #STANDARD DEVIATION
> #(a) Direct Method
> SD_direct <- sqrt(sum(f * (x - mean_x)^2) / sum(f))
> SD_direct
[1] 1.313155
> #(b) Assumed Mean Method
> SD_assumed <- sqrt(sum(f * d^2) / sum(f) - (sum(f * d) /
+ sum(f))^2)
> SD_assumed
[1] 1.313155
> #(c) Step Deviation Method
> SD_step <- sqrt(sum(f * u^2) / sum(f) - (sum(f * u) /
+ sum(f))^2) * h
> SD_step
[1] 1.313155
> |
```

OUTPUT:-

Environment	History	Connections	Tutorial
Import Dataset	12 MiB		
R	Global Environment		
Values			
A	2		
coef_MD_mean	0.510493827160494		
coef_MD_median	0.5125		
coef_QD	0.5		
coef_range	1		
d	num [1:6] -2 -1 0 1 2 3		
data	num [1:40] 0 0 0 0 1 1 1 1 1 1 ...		
f	num [1:6] 4 12 11 7 4 2		
freq	'table' int [1:11(1d)] 1 1 3 2 3 5 2 2 3 2 ...		
h	1		
MD_mean	1.03375		
MD_median	1.025		
mean_assumed	2.025		
mean_score	15.12		
mean_step	2.025		
mean_x	2.025		
median_score	15		
median_x	2		
mode_score	15		
mode_x	1		
n	25L		
Q1	1		
Q3	3		
QD	1		
range_score	10	3	
range_x	5		
scores	num [1:25] 12 15 18 14 16 12 19 15 13 17 ...		
SD_assumed	1.31315459866689		
SD_direct	1.31315459866689		
SD_step	1.31315459866689		
u	num [1:6] -2 -1 0 1 2 3		
Files	Plots	Packages	Help Viewer Presentation

Que -2(code part):-

#Question 2:

```
x <- c(0,1,2,3,4,5)
```

```
f <- c(4,12,11,7,4,2)
```

```
data <- rep(x, f)
```

```
length(data)
```

```
#MEAN
```

```
 #(A)DIRECT METHOD
```

```

mean_x <- sum(x * f) / sum(f)
mean_x
#(b) Assumed Mean Method
A <- 2 # assumed mean
d <- x - A
mean_assumed <- A + sum(f * d) / sum(f)
mean_assumed
#(c) Step Deviation Method
h <- 1
u <- (x - A) / h
mean_step <- A + (sum(f * u) / sum(f)) * h
mean_step
#MEDIAN
median_x <- median(data)
median_x
#Mode
mode_x <- x[f == max(f)]
mode_x
#RANGE & COEFFICIENT OF RANGE
range_x <- max(x) - min(x)
range_xcoef_range <- (max(x) - min(x)) / (max(x) + min(x))
coef_range
#QUARTILE DEVIATION (Q.D.) & COEFFICIENT
Q1 <- as.numeric(quantile(data, 0.25))
Q3 <- as.numeric(quantile(data, 0.75))
QD <- (Q3 - Q1) / 2
QD
coef_QD <- (Q3 - Q1) / (Q3 + Q1)
coef_QD
#MEAN DEVIATION ABOUT MEAN & COEFFICIENT
MD_mean <- sum(abs(data - mean_x)) / length(data)
MD_mean
coef_MD_mean <- MD_mean / mean_x
coef_MD_mean

```

```

#MEAN DEVIATION ABOUT MEDIAN & COEFFICIENT
MD_median <- sum(abs(data - median_x)) / length(data)
MD_median
coef_MD_median <- MD_median / median_x
coef_MD_median
#STANDARD DEVIATION
#(a) Direct Method
SD_direct <- sqrt(sum(f * (x - mean_x)^2) / sum(f))
SD_direct
#(b) Assumed Mean Method
SD_assumed <- sqrt(sum(f * d^2) / sum(f) - (sum(f * d) /
sum(f))^2)
SD_assumed
#(c) Step Deviation Method
SD_step <- sqrt(sum(f * u^2) / sum(f) - (sum(f * u) /
sum(f))^2) * h
SD_step

```

QUESTION 3:-

3. A housing society recorded the monthly electricity consumption (in kilowatt-hours, kWh) of 150 households to analyse energy usage patterns. The data is summarized in the table below:

Consumption (kWh)	100-150	150-200	200-250	250-300	300-350	350-400	400-450
No. of Households (f)	12	28	45	35	18	10	2

CODE:-

```
DA1.R*
Source on Save
Run
Source

1 #question 3:
2 # Class intervals
3 lower <- c(100,150,200,250,300,350,400)
4 upper <- c(150,200,250,300,350,400,450)
5 # Frequencies
6 f <- c(12,28,45,35,18,10,2)
7 # Class mid-points
8 mid <- (lower + upper) / 2
9 sum(f) # total households
10 #Mean
11 #(a) Direct Method
12 mean_x <- sum(mid * f) / sum(f)
13 mean_x
14 #(b) Assumed Mean Method
15 A <- 225 # assumed mean
16 d <- mid - A
17 mean_assumed <- A + sum(f * d) / sum(f)
18 mean_assumed
19 #(c) Step Deviation Method
20 h <- 50
21 u <- (mid - A) / h
22 mean_step <- A + (sum(f * u) / sum(f)) * h
23 mean_step
24 #Median
25 cf <- cumsum(f)
26 cf
27 N <- sum(f)
28 median_class <- which(cf >= N/2)[1]
29 l <- lower[median_class]
30 c_prev <- ifelse(median_class == 1, 0, cf[median_class -
31 1])
32 f_m <- f[median_class]
33 median_x <- l + ((N/2 - c_prev) / f_m) * h
34 median_x
35 #Mode
36 modal_class <- which.max(f)
37 l1 <- lower[modal_class]
38 f1 <- f[modal_class]
39 f0 <- ifelse(modal_class == 1, 0, f[modal_class - 1])

88:8 (Top Level)
R Script
```

Console

CONSOLE:-

```
Source
Console Terminal Background Jobs
R 4.5.2 · ~/new da/
> #Mode
> mode_x <- x[f == max(f)]
> mode_x
[1] 1
> #RANGE & COEFFICIENT OF RANGE
> range_x <- max(x) - min(x)
> range_x
[1] 5
> coef_range <- (max(x) - min(x)) / (max(x) + min(x))
> coef_range
[1] 1
> #QUARTILE DEVIATION (Q.D.) & COEFFICIENT
> Q1 <- as.numeric(quantile(data, 0.25))
> Q3 <- as.numeric(quantile(data, 0.75))
> QD <- (Q3 - Q1) / 2
> QD
[1] 1
> coef_QD <- (Q3 - Q1) / (Q3 + Q1)
> coef_QD
[1] 0.5
> #MEAN DEVIATION ABOUT MEAN & COEFFICIENT
> MD_mean <- sum(abs(data - mean_x)) / length(data)
> MD_mean
[1] 1.03375
> coef_MD_mean <- MD_mean / mean_x
> coef_MD_mean
[1] 0.5104938
> #MEAN DEVIATION ABOUT MEDIAN & COEFFICIENT
> MD_median <- sum(abs(data - median_x)) / length(data)
> MD_median
[1] 1.025
> coef_MD_median <- MD_median / median_x
> coef_MD_median
[1] 0.5125
> #STANDARD DEVIATION
> #(a) Direct Method
> SD_direct <- sqrt(sum(f * (x - mean_x)^2) / sum(f))
> SD_direct
[1] 1.313155
> #(b) Assumed Mean Method
```

```
Source
Console Terminal × Background Jobs ×
R 4.5.2 · ~/new da/
> l3 <- lower[Q3_class]
> c3 <- cf[Q3_class - 1]
> f3 <- f[Q3_class]
> Q3 <- l3 + ((3*N/4 - c3) / f3) * h
> QD <- (Q3 - Q1) / 2
> QD
[1] 46.875
> coef_QD <- (Q3 - Q1) / (Q3 + Q1)
> coef_QD
[1] 0.1933702
> #MEAN DEVIATION ABOUT MEAN & COEFFICIENT
> MD_mean <- sum(f * abs(mid - mean_x)) / sum(f)
> MD_mean
[1] 56.2
> coef_MD_mean <- MD_mean / mean_x
> coef_MD_mean
[1] 0.2303279
> #MEAN DEVIATION ABOUT MEDIAN & COEFFICIENT
> MD_median <- sum(f * abs(mid - median_x)) / sum(f)
> MD_median
[1] 55.51852
> coef_MD_median <- MD_median / median_x
> coef_MD_median
[1] 0.2324031
> #STANDARD DEVIATION
> #(a) Direct Method
> SD_direct <- sqrt(sum(f * (mid - mean_x)^2) / sum(f))
> SD_direct
[1] 68.71924
> #(b) Assumed Mean Method
> SD_assumed <- sqrt(sum(f * d^2) / sum(f) - (sum(f * d) /
+ sum(f))^2)
> SD_assumed
[1] 68.71924
> #(c) Step Deviation Method
> SD_step <- sqrt(sum(f * u^2) / sum(f) - (sum(f * u) /
+ sum(f))^2) * h
> SD_step
[1] 68.71924
```

OUTPUT:-

Environment

History

Connections

Tutorial

Import Dataset

16 MiB

List

R

Global Environment

lower	num [1:7] 100 150 200 250 300 350 400
MD_mean	56.2
MD_median	55.5185185185185
mean_assumed	244
mean_score	15.12
mean_step	244
mean_x	244
median_class	3L
median_score	15
median_x	238.888888888889
mid	num [1:7] 125 175 225 275 325 375 425
modal_class	3L
mode_score	15
mode_x	231.481481481481
n	25L
N	150
Q1	195.535714285714
Q1_class	2L
Q3	289.285714285714
Q3_class	4L
QD	46.875
range_score	10
range_x	range_score (numeric , 56 bytes)
scores	num [1:25] 12 15 18 14 16 12 19 15 13 17 ...
SD_assumed	68.7192355409556
SD_direct	68.7192355409556
SD_step	68.7192355409556
u	num [1:7] -2 -1 0 1 2 3 4
upper	num [1:7] 150 200 250 300 350 400 450
x	num [1:6] 0 1 2 3 4 5

Functions

mode_function

function (x)

Files

Plots

Packages

Help

Viewer

Presentation

Que-3(code part):-

#question 3:

Class intervals

lower <- c(100,150,200,250,300,350,400)

upper <- c(150,200,250,300,350,400,450)

Frequencies

f <- c(12,28,45,35,18,10,2)

Class mid-points

```

mid <- (lower + upper) / 2
sum(f) # total households
#Mean
#(a) Direct Method
mean_x <- sum(mid * f) / sum(f)
mean_x
#(b) Assumed Mean Method
A <- 225 # assumed mean
d <- mid - A
mean_assumed <- A + sum(f * d) / sum(f)
mean_assumed
#(c) Step Deviation Method
h <- 50
u <- (mid - A) / h
mean_step <- A + (sum(f * u) / sum(f)) * h
mean_step
#Median
cf <- cumsum(f)
cf
N <- sum(f)
median_class <- which(cf >= N/2)[1]
l <- lower[median_class] c_prev <- ifelse(median_class == 1, 0,
cf[median_class -
1])
f_m <- f[median_class]
median_x <- l + ((N/2 - c_prev) / f_m) * h
median_x
#Mode
modal_class <- which.max(f)
l1 <- lower[modal_class]
f1 <- f[modal_class]
f0 <- ifelse(modal_class == 1, 0, f[modal_class - 1])
f2 <- ifelse(modal_class == length(f), 0, f[modal_class +
1])

```

```

mode_x <- l1 + ((f1 - f0) / (2*f1 - f0 - f2)) * h
mode_x
#RANGE & COEFFICIENT OF RANGE
range_x <- max(upper) - min(lower)
range_x
coef_range <- (max(upper) - min(lower)) / (max(upper) +
min(lower))
coef_range
#QUARTILE DEVIATION (Q.D.) & COEFFICIENT
# Q1
Q1_class <- which(cf >= N/4)[1]
l1 <- lower[Q1_class]
c1 <- ifelse(Q1_class == 1, 0, cf[Q1_class - 1])
f1 <- f[Q1_class]
Q1 <- l1 + ((N/4 - c1) / f1) * h
# Q3
Q3_class <- which(cf >= 3*N/4)[1]
l3 <- lower[Q3_class]
c3 <- cf[Q3_class - 1]f3 <- f[Q3_class]
Q3 <- l3 + ((3*N/4 - c3) / f3) * h
QD <- (Q3 - Q1) / 2
QD
coef_QD <- (Q3 - Q1) / (Q3 + Q1)
coef_QD
#MEAN DEVIATION ABOUT MEAN & COEFFICIENT
MD_mean <- sum(f * abs(mid - mean_x)) / sum(f)
MD_mean
coef_MD_mean <- MD_mean / mean_x
coef_MD_mean
#MEAN DEVIATION ABOUT MEDIAN & COEFFICIEN
MD_median <- sum(f * abs(mid - median_x)) / sum(f)
MD_median
coef_MD_median <- MD_median / median_x
coef_MD_median

```

#STANDARD DEVIATION

#(a) Direct Method

```
SD_direct <- sqrt(sum(f * (mid - mean_x)^2) / sum(f))
```

```
SD_direct
```

#(b) Assumed Mean Method

```
SD_assumed <- sqrt(sum(f * d^2) / sum(f) - (sum(f * d) /  
sum(f))^2)
```

```
SD_assumed
```

#(c) Step Deviation Method

```
SD_step <- sqrt(sum(f * u^2) / sum(f) - (sum(f * u) /  
sum(f))^2) * h
```

```
SD_step
```